Operational Research: Tutorial 5

Mixed Integer Programming

The more complicated questions on the tutorial are marked by:

* Medium

** Hard

Exercises

1) Table 1 gives weights and survival values for items that can be chosen to take with you to a desert island. You can carry 15kg.
   a. Maximise the survival value of your selection.

   w_i – weight of item i (kg)

   s_i – survival value of item i

   x_i – number of times item i is packed (0 or 1 in a and c; any whole number in b)

Maximise

$$\sum_{i=1}^{8} s_i x_i$$

subject to:

$$\sum_{i} w_i x_i \leq 15$$

and

$$x_i \geq 0$$

$$x_i \leq 1$$

$$x_i \in \mathbb{Z}$$

   b. Assume that you can pick an item more than once. (This would probably be a bad idea in practice!)

Maximise

$$\sum_{i=1}^{8} s_i x_i$$

subject to:

$$\sum_{i} w_i x_i \leq 15$$

and

$$x_i \geq 0$$

Sensitivity: Internal

$$x_i \in \mathbb{Z}$$

c. Ignoring b), assume that you must pick at least 5 items.

Item	Weight (kg)	Survival Value
1	2	10
2	3	50
3	2	40
4	6	100
5	6	100
6	8	130
7	3	70
8	5	80

Table 1

Maximise

$$\sum_{i=1}^{8} s_i x_i$$

subject to:

$$\sum_i w_i x_i \leq 15$$

$$\sum_i x_i \geq 5$$

and

$$x_i \geq 0$$

$$x_i \leq 1$$

$$x_i \in \mathbb{Z}$$

The optimal solutions for the three MILP models are given below. Please look at the code to see the implementations.

Sensitivity: Internal

```
sol =

     0
     0
     0
     1
     1
     0
     1
     0


fval =

   -270
```

```
sol =

     0
     0
     0
     0
     0
     0
     0
     5
     0


fval =

   -350
```

```
sol =

     1
     1
     1
     0
     0
     0
     1
     1


fval =

   -250
```

2) Table 2 gives the cities that would be safely covered by a fire station located at the given city. Formulate a set covering problem to minimise the number of fire stations that require building so that every city is safely covered. In which cities should the fire stations be built?

City	Covers
1	1,2,3
2	1,2,4
3	1,3
4	2,4,5
5	4,5,6
6	5,6,7
7	6,7

Table 2

minimise

$$\sum_i x_i$$

subject to:

$$x_1 + x_2 + x_3 \geq 1 \quad \text{(City 1 Constraint)}$$

$$x_1 + x_2 + x_4 \geq 1 \quad \text{(City 2 Constraint)}$$

$$x_1 + x_3 \geq 1 \quad \text{(City 3 Constraint)}$$

$$x_2 + x_4 + x_5 \geq 1 \quad \text{(City 4 Constraint)}$$

$$x_4 + x_5 + x_6 \geq 1 \quad \text{(City 5 Constraint)}$$

$$x_5 + x_6 + x_7 \geq 1 \quad \text{(City 6 Constraint)}$$

$$x_6 + x_7 \geq 1 \quad \text{(City 7 Constraint)}$$

and

$$x_i \in \{0,1\}, \ \forall i$$

Sensitivity: Internal

```
sol =

     0
     0
     1
     1
     0
     0
     1


fval =

     3
```

Note you might get a different selection, but the minimum will be 3!

3) *C&S* (Management Consultants) have been employed to help the *Icki-Toffee* Company to develop a popular, economical but more health-conscious selection of candies to add to their scrumptious range of toffees. There has been much publicity that the Government are so worried about obesity that they plan to introduce a coding system for all food products which will inform the public about products containing foods which may be adding to the problem. There are rumours that the Government aim to strongly promote the concept of fruit as a healthy alternative to sweets, so *Icki-Toffee* are very keen to introduce fruit, in the form of fillings, soft centres etc into their range. They are determined to ensure that at least one of their selections of candies does not fall into the worst category.

   - The selection chosen must weigh at least 320g
   - Total fat content by weight must be at most 45% of the total weight of the selection.
   - Calorific value of the selection must be at most 2850 calories.
   - Total weight is limited to 1.5 times the minimum weight of 320g

   The quantities of fat, calories, weight and cost for each candy are given in Table 3.

   Minimise the total cost subject to the following: **

   - Ensure that your selection has at least 6 different candies
   - Ensure your selection includes at least 3 of each of the selected candies.
   - Limit selected candies to a maximum of 6 each in your selection

Candy - x_i	**Fat (g) - F_i**	**Calories – K_i**	**Weight (g) - W_i**	**Cost - C_i**
Buttery butterscotch	20	100	24	12
Caramel softie	16	100	22	12
Orange dream	10	80	15	12
Coconut Ice	7	50	12	15
Lemon sorbet	6	60	10	13
Lime sherbet	5	40	10	13
Strawberry surprise	13	80	14	16
Raspberry mousse	15	90	17	16
Ginger toffee	18	80	20	15
Rose-water posy	-	145	8	10

Sensitivity: Internal

Walnut whirl	19	120	20	18
Grape bunch	-	120	15	20

Table 3: Table to show content information about each candy

x_i – number of item i selected

$$\delta_i = \begin{cases} 1 - item\ i\ selected \\ 0 - \ item\ i\ not\ selected \end{cases}$$

minimise

$$\sum_i C_i x_i$$

subject to:

$$\sum_i W_i x_i \geq 320$$

$$\sum_i W_i x_i \leq 480$$

$$\sum_i K_i x_i \leq 2850$$

$$\sum_i F_i x_i \leq 0.45 \sum_i W_i x_i$$

$$\sum_i \delta_i \geq 6$$

$$x_i \geq 3\delta_i$$

$$x_i \leq 6\delta_i$$

and

$$x_i \geq 0, \ \forall i$$

$$x_i \in \mathbb{Z}, \quad \forall i$$

$$\delta_i \in \{0,1\}, \quad \forall i$$

Check out the code for this, the left column below is the x_i values and the right column is the δ_i values

```
ans =

     3
     3
     0
     0
     3
     3
     0
     0
     0
     4
     0
     6
```

```
ans =

     1
     1
     0
     0
     1
     1
     0
     0
     0
     1
     0
     1
```

```
fval =

   310
```

Note: this problem has more than one cheapest selection, so your code may show a different mix from the one above. Any selection that costs 310 and meets the constraints is correct.

Sensitivity: Internal